

Automatic Handwriting Style Classification: Cursive, Printed and Mixed

Technical Analysis Document — Computer Vision Exam Project

1. Problem Statement

This project aims to train a model capable of distinguishing handwritten text based on the type of characters used, classifying it into cursive, printed, or mixed. This work can represent an initial phase for the development of broader systems for the digitisation of handwritten texts, adaptive OCR that selects the most appropriate recognition strategy based on the detected writing style, or accessibility tools for transcribing handwritten documents.

The three classes refer to the type of characters used in writing. The "cursive" class collects scripts where there is a substantial connection between letters, with continuous strokes linking one letter to the next. The "printed" class refers to sentences written in lowercase printed text, with separate letters and isolated strokes. The third class, "mixed", represents an intermediate area that collects handwriting styles that, in their personalisation, exhibit characteristics of both styles: for example, printed text with occasional ligatures, or cursive with frequent pen lifts (sometimes called "script").

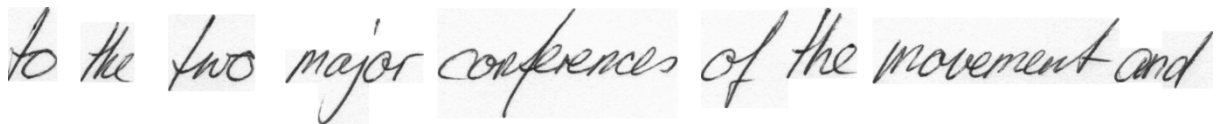
Handwriting classification is not a trivial problem for two main reasons. The first is the strong inter-writer variability: different people express the same style with very different degrees of connection, inclination and regularity, making it difficult to identify universal geometric rules. The second reason concerns the nature of the "mixed" class itself, which by definition does not possess distinctive characteristics of its own, but is defined by exclusion relative to the other two classes: mixed writing shares properties of both cursive and printed text, making separation intrinsically more ambiguous, as confirmed by the results of this project.

2. Dataset

The dataset source is the **IAM Handwriting Database** (<https://fki.tic.heia-fr.ch/databases/iam-handwriting-database>). Labelling into the three classes was performed manually, visually evaluating for each line of writing the presence of letter connections and the type of characters used.

The final dataset consists of 147 images per class, for a total of 441 samples; each image represents a single line of handwritten text. The number of samples per class was balanced — choosing 147 as the limit imposed by the least numerous class, "mixed" (to prevent the model from developing a bias towards the most represented class, learning to favour it regardless of the actual writing characteristics).

Writing Examples



Example of cursive handwriting

The image shows a sample of handwritten text in a lowercase printed style. The text is "borough Labour Party Conference I am". The letters are somewhat irregular and slanted, typical of a printed style written by hand.

Example of lowercase printed handwriting

The image shows a sample of handwritten text in a mixed style. The text is "Hounslow. Miss Gump had been the housekeeper at the". The letters are more cursive and connected than the printed example, but still somewhat legible.

Example of handwriting classified as "mixed"

3. Methodology

3.1 Preprocessing

Each image is converted to grayscale and subsequently binarised using the Otsu method, which automatically calculates the optimal threshold by maximising the inter-class variance between ink and background. The binary inversion (ink = 255, background = 0) is due to a convention ensuring that OpenCV functions for connected component analysis (`cv2.connectedComponentsWithStats`, `cv2.findContours`) correctly treat the ink as a foreground object.

For each image, two distinct versions of the binary are saved: a raw version (`_otsu.png`) and a version subjected to morphological opening with a 2×2 rectangular kernel (`_clean.png`). This architectural choice arises from the fact that morphological opening (while useful for removing point noise) tends to erode the thin strokes that close letter loops, artificially reducing the number of detectable rings. The `_otsu.png` version is therefore used for the loop density feature, while the `_clean.png` version is used for features based on connected component segmentation (continuity, average width, distance variance).

3.2 Feature Extraction

As anticipated in the introduction, the extracted features do not concern the shape of individual letters in the different writing styles, but are the result of a global overview of writing dictated by the continuity of the graphic gesture, a more comprehensive vision typical, for example, of Italian graphology. Class names and function names are intentionally kept in Italian to emphasise this approach.

Five features were designed and extracted: four scalar features based on connected components and one vectorial feature (HOG). For each scalar feature, statistical validation was conducted on the entire dataset ($n=147$ per class), computing mean and standard deviation to verify the consistency of the expected ordering among the three classes.

- Feature 1 — Component density (continuity/ligature): number of connected components normalised by line width. Mean values: cursive 13.05, mixed 17.71, printed 22.26. Low density indicates letters fused into continuous strokes (cursive); high density indicates isolated letters (printed). This is the key feature for class separation.
- Feature 2 — Loop density: number of closed loops (topological holes inside components, detected via the hierarchy returned by `cv2.findContours` with `RETR_CCOMP` mode) normalised by line width. Mean values: cursive 8.47, mixed 8.35, printed 7.51. The signal is

in the expected direction but weak, with strong overlap between cursive and mixed distributions. This feature was chosen because cursive writing typically exhibits more loops, given the shape of its letters.

- Feature 3 — Average blob width: mean width of the bounding box of connected components, normalised by line width. Mean values: cursive 41.29, mixed 30.66, printed 22.88. The choice reflects the fact that the less frequent letter separations in cursive writing produce wider clusters of letters.
- Feature 4 — Inter-component distance variance: variance of horizontal distances between consecutive component centroids. Mean values: cursive 3591.67, mixed 1780.21, printed 930.63. Measures spacing regularity: printed text has more uniform spacing between isolated letters, while cursive varies more (linked strokes alternating with wide gaps between words).
- Feature 5 — HOG (Histogram of Oriented Gradients): computed on images resized to 256×64 pixels, with 8 orientations and 16×16 cells, producing a final vector of 512 dimensions per image.

The final feature vector for each image is therefore 516-dimensional (4 scalar features + 512 HOG components).

3.3 Classical Classifier — Random Forest

A Random Forest was chosen as the classical classification model, compared in a preliminary phase with a Support Vector Machine with RBF kernel. The Random Forest was preferred for its greater robustness on small datasets and for the interpretability of the model.

Features were standardised with StandardScaler before training. Evaluation was conducted via StratifiedKfold with 5 folds, to obtain a more stable estimate of metrics compared to a single train/test split. Cross-validation results are reported in Section 4.

3.4 Deep Learning Approach — Transfer Learning Evaluation

With a dataset of only 441 samples, training a deep neural network from scratch would almost certainly have led to overfitting: for this reason, a transfer learning approach was tested using MobileNetV2 pre-trained on ImageNet, which limits the number of parameters to learn by leveraging visual features already acquired on millions of images.

Contrary to the common expectation that a deep learning approach always outperforms a classical one, on this dataset the Random Forest significantly outperforms MobileNetV2 (46%). The three main causes are analysed in Section 5.

3.5 Post-processing — Confidence Thresholding

The Random Forest returns, for each classified image, not only the predicted label but also a probability for each of the three classes via `predict_proba`. Post-processing consists of applying a minimum confidence threshold to these probabilities: if the maximum probability is below 40%, the prediction is marked as "uncertain" rather than forcing a potentially incorrect label.

The 40% threshold was chosen empirically by testing values between 0.35 and 0.50. With threshold 0.35, no sample is discarded (too permissive), while with threshold 0.50, 56 out of 89 samples are rejected (too restrictive for a 3-class problem, where random chance probability per class is approximately 0.33). The 0.40 threshold represents the most useful compromise: it discards 15 out of 89 samples, of which 8 belong to the mixed class (53% of uncertain cases), confirming that the model is structurally less confident on this class.

4. Experimental Results

4.1 Classical Classifier Results — Random Forest

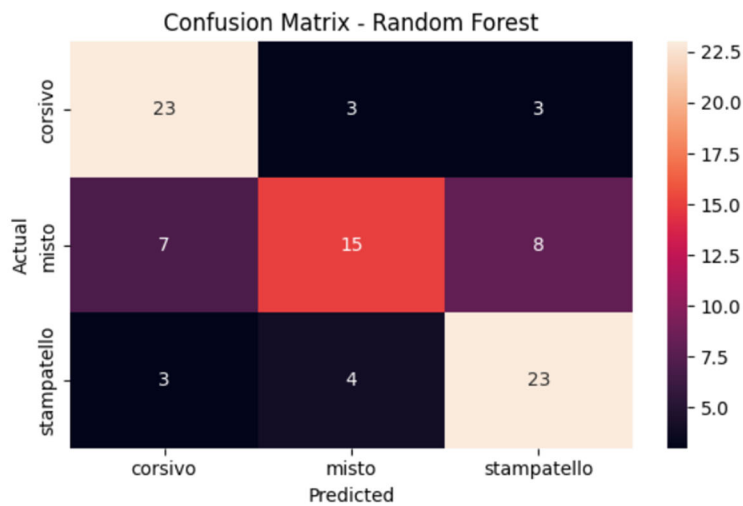
Model	Accuracy	F1 macro
Random Forest (5-fold cross-validation)	64,4% ($\pm$ 5.3%)	63,5% ($\pm$ 5.4%)

The following metrics refer to evaluation on the held-out test set (20% of the dataset, 89 samples):

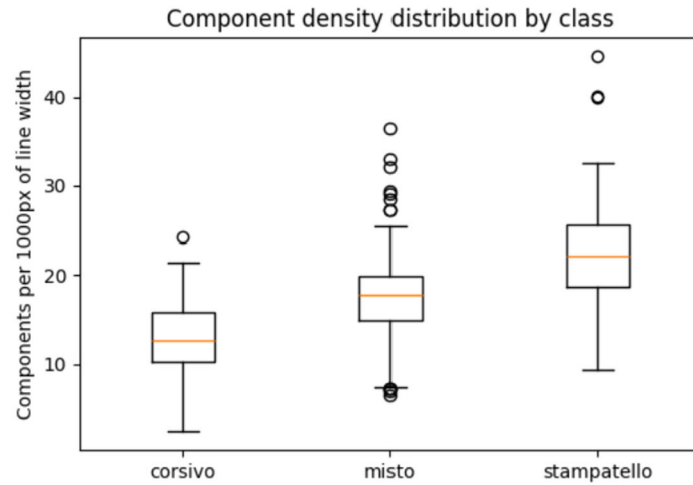
- Cursive: precision 0.70, recall 0.79, F1 0.74 (support 29)
- Mixed: precision 0.68, recall 0.50, F1 0.58 (support 30)
- Printed: precision 0.68, recall 0.77, F1 0.72 (support 30)

The error pattern confirms that cursive and printed are correctly distinguished in most cases, while the mixed class is the most error-prone, with errors distributed roughly equally towards the other two classes (discussed in Section 5).

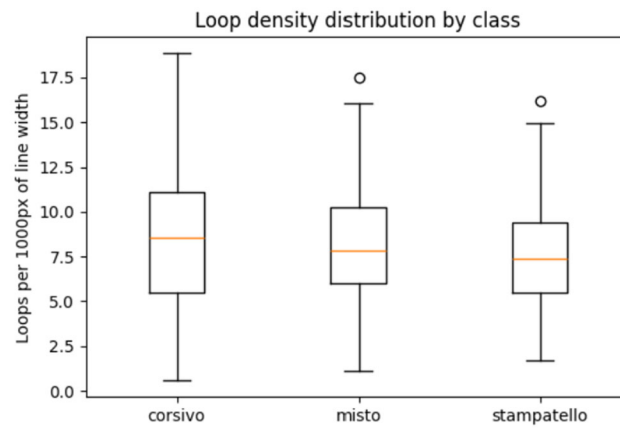
	precision	recall	f1-score	support
corsivo	0.70	0.79	0.74	29
misto	0.68	0.50	0.58	30
stampatello	0.68	0.77	0.72	30
accuracy			0.69	89
macro avg	0.69	0.69	0.68	89
weighted avg	0.68	0.69	0.68	89



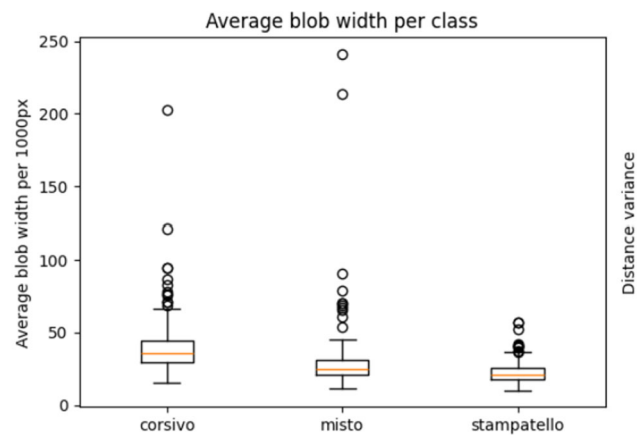
Classification report and confusion matrix — Random Forest



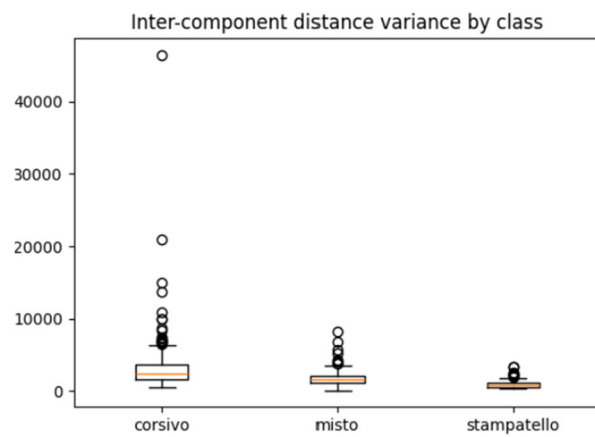
Component density distribution by class (Feature 1)



Loop density distribution by class (Feature 2)



Average blob width distribution by class (Feature 3)

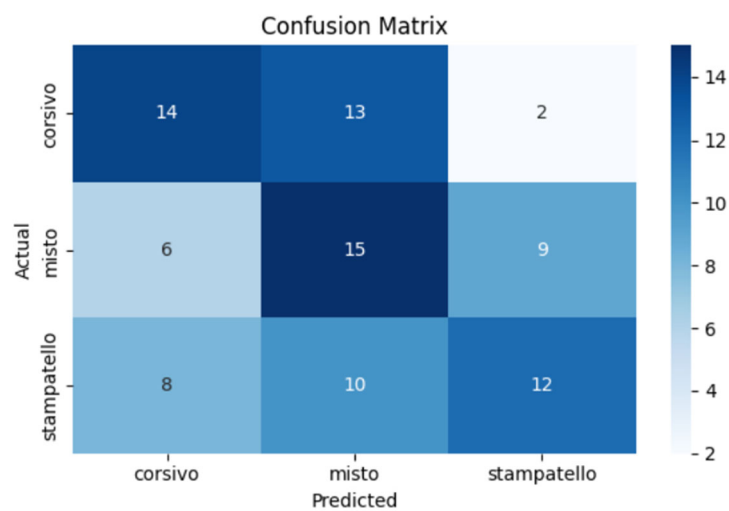


Inter-component distance variance by class (Feature 4)

4.2 Transfer Learning Experiment

An experiment was conducted with MobileNetV2 pre-trained on ImageNet, adapted to the problem via transfer learning. The model achieved an accuracy of 46% on the validation set, a result inferior to the classical classifier. The reasons for this outcome are analysed in Section 5.

	precision	recall	f1-score	support
corsivo	0.50	0.48	0.49	29
misto	0.39	0.50	0.44	30
stamatello	0.52	0.40	0.45	30
accuracy			0.46	89
macro avg	0.47	0.46	0.46	89
weighted avg	0.47	0.46	0.46	89



Classification report and confusion matrix — MobileNetV2

5. Failure Analysis

5.1 Confusion of the "Mixed" Class

The mixed class has the lowest recall (50%), with errors distributed roughly equally between cursive and printed (7 and 8 samples respectively out of 30 misclassified in the validation split). This result was expected from the feature validation phase: for three of the four scalar features (component density, average width, distance variance), the mean values of the mixed class fall in an intermediate position between cursive and printed, often with overlapping ranges relative to standard deviations. This confirms that the mixed class, being defined by exclusion relative to the other two, does not possess distinctive geometric characteristics of its own, but "intermediate" values by construction. This ambiguity also emerges in the analysis of model confidences: applying a 40% threshold on the maximum predicted probability, the samples rejected as "uncertain" belong for 53% to the mixed class, confirming that the model internally recognises the difficulty of this class.

5.2 Deep Learning Model

With a dataset of only 441 samples, training a deep neural network from scratch would almost certainly have led to overfitting. For this reason, a transfer learning approach was tested with MobileNetV2 pre-trained on ImageNet. Contrary to the common expectation that a deep learning approach always outperforms a classical one, on this dataset the Random Forest significantly outperforms MobileNetV2. Three main causes were identified:

- Dataset size: 441 samples (353 for training) are insufficient to effectively adapt even just the classification head of a pre-trained network, with a concrete risk of overfitting despite freezing the base weights.
- Domain mismatch: MobileNetV2 is pre-trained on ImageNet, a dataset of natural colour photographs with rich textures. The project images are instead binary (ink/background) and artificially converted to 3 identical RGB channels, a domain very distant from the pre-training domain.
- Handcrafted features are task-specific: the four scalar features designed directly capture the structural properties that define writing style (continuity, loop shape, spacing), while the network must attempt to relearn these properties from raw pixels with insufficient data.

6. Ethical Considerations

Handwriting can function as a biometric identifier, analogously to a fingerprint or voice. A handwriting classification or recognition system, if applied to real documents, must handle authors' data with particular attention to privacy.

Moreover, the IAM dataset was collected from a limited group of writers in terms of language and origin. A model trained on this data may generalise poorly to writing styles not represented in the training set (e.g. left-handed writers, writers from other countries, or people with motor disabilities affecting handwriting). Using such a system on populations different from the training population would require a new validation phase.